

Problem 15.27

Susan and Jeff each make deposits of 100 at the end of each year for 40 years. Starting at the end of the 41st year, Susan makes annual withdrawals of X for 15 years and Jeff makes annual withdrawals of Y for 15 years. Both funds have a balance of 0 after the last withdrawal. Susan's fund earns an annual effective interest rate of 8%. Jeff's fund earns an annual effective interest rate of 10%. Calculate $Y - X$.

Solution

Each person makes deposits for 40 years and then makes withdrawals for 15 years. The accumulated value at time 40 of the deposits must equal the present value at time 40 of the withdrawals.

Susan

Susan earns 8% effective annually. Her deposits accumulate to time 40 as

$$100s_{\overline{40}|}.$$

The withdrawals of X from times 41 through 55 have present value at time 40

$$Xa_{\overline{15}|}.$$

Thus,

$$100s_{\overline{40}|} = Xa_{\overline{15}|}.$$

Using $i = 0.08$,

$$\begin{aligned} X &= \frac{100s_{\overline{40}|}}{a_{\overline{15}|}} \\ &= \frac{100 \left(\frac{(1.08)^{40} - 1}{0.08} \right)}{\frac{1 - (1.08)^{-15}}{0.08}} \\ &\approx 3026.5455. \end{aligned}$$

Jeff

Jeff earns 10% effective annually. Similarly,

$$100s_{\overline{40}|} = Ya_{\overline{15}|}.$$

Using $i = 0.10$,

$$\begin{aligned} Y &= \frac{100s_{\overline{40}|}}{a_{\overline{15}|}} \\ &= \frac{100 \left(\frac{(1.10)^{40} - 1}{0.10} \right)}{\frac{1 - (1.10)^{-15}}{0.10}} \\ &\approx 5818.9315. \end{aligned}$$

Therefore,

$$\begin{aligned} Y - X &\approx 5818.9315 - 3026.5455 \\ &\approx 2792.3860. \end{aligned}$$

Hence,

$$\boxed{Y - X \approx 2792.39}.$$